

An LLM Agent for the Selection, Validation and Interpretation of Statistical Methods

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1. Purpose and summary

This report records the state of the project at the week-four freeze gate — the point the plan defines as the decision point before the held-out evaluation is executed. It covers what has been built, what the development-set evidence shows, which risks have materialised, and what remains.

Summary. The system, the benchmark and the evaluation harness are complete and the freeze conditions are met. The development-set evidence supports the central hypothesis, with an uplift well above the ten-point target. The held-out evaluation is the remaining work. One planned measurement has failed and is reported as failed rather than quietly dropped.

2. Progress against the acceptance gates

Week	Gate	Status
1	Raw data reproducible; development cases pass validation	Met. Four UCI sources acquired with checksums and a recorded manifest; development cases validate.
2	Every method unit-tested; all cases valid; labels resolved	Met. All 64 cases validate, 12 of them abstention cases; a full label audit passes.
3	All three systems finish every development case; no untraced claim	Met. Every arm completes the development partition and every rendered section resolves to a recorded tool output.
4	Development accuracy $\geq 80\%$; fidelity $\geq 98\%$; candidate tagged	Met. Structured arm at 93.8% on the development partition; provenance enforced at schema level; prompt, model, code and runner frozen.
5	Held-out results reproducible from raw logs	Outstanding. The runner, its execution guard and the pre-flight are in place and tested; execution is the remaining step.

Table 1. Status against the planned acceptance gates.

3. What has been built

Fourteen statistical methods with effect sizes and confidence intervals, five diagnostic tools, and a deterministic state machine around one tool-calling model. Validated libraries perform every numerical

operation; no statistic is reimplemented.

The three arms share one core. Every driver imports the same tool registry, result store and report schema, so the fairness of the baselines is a property of the repository rather than a claim in prose. The provenance contract is enforced at schema level: the report format rejects raw numeric literals in prose, and a reference to a result that was never computed fails the run.

4. Benchmark status

64 validated case packages, 12 of which have no admissible in-scope method. Cases are generated from a declarative registry, so the benchmark regenerates from source rather than existing as a set of files that must be preserved.

A label audit was run over the full set, comparing each gold label against the realised sample rather than against the generator's intent. It caught a routing defect in which extreme skew was assessed after a variance check rather than before, which had mislabelled a public case. The audit now passes on every case and runs as part of the standard verification.

5. Preliminary results

These are **development-set** results and must not be presented as the headline experiment: prompts were developed against these cases, so the numbers are optimistic by construction. They are reported here because they are what the freeze decision rests on.

System	n	Accuracy	95% CI (Wilson)	Abstention recall	Unsafe rate
A — Direct	16	0.875	[0.640, 0.965]	0.33	0.125
B — Tools	16	0.500	[0.280, 0.720]	0.00	0.188
C — Protocol	16	0.938	[0.717, 0.989]	1.00	0.000

Table 2. Development partition, structured effort level. Paired comparison of the tool baseline against the protocol gives an uplift of +0.438 against a ten-point target.

The sharpest separation is abstention. On the design-hazard cases the structured arm declined every time and the tool-enabled baseline never did, fitting a plausible-looking model to data whose independence assumption fails. That is precisely the failure the project exists to prevent, and it is the result that most needs held-out confirmation.

6. Offline calibration

A deterministic policy client was run through the same harness at two competence levels, 432 runs each, to validate the harness independently of any model. The expert policy scores perfectly, which is *circular* — that policy derived the gold labels — and is reported as such rather than as a result.

The informative row is the naive policy. Inside the identical state machine it reaches 0.604 with zero abstention recall. This demonstrates that the architecture is not what produces the uplift — the decision policy inside it is — and it is the strongest available evidence that the harness measures the intended quantity.

7. Deviations recorded so far

Deviation	Reason
temperature: 0 is not sent	The parameter has been removed on current models and returns an error. Determinism is controlled by a pinned model, fixed effort, a frozen prompt hash and measured repetition variance.
Abstention cases raised from 8 to 12	Eight cases could not support an interval narrow enough to distinguish the arms on the safety-critical metric.
Numeric fidelity is prevented, not verified	Schema-level provenance makes fabrication impossible in the structured arm, which converts a metric into a guarantee. Reported as such rather than as a score.
The verifier moved from week four to week two	It changes what the structured arm produces, so building it after the development runs would have invalidated them.
Not every method appears in both partitions	Twelve synthetic development slots cannot cover fourteen methods plus abstention. Four methods appear only in the held-out set — a cleaner generalisation test, but an unplanned one.
Synthetic gold derives from the realised sample	A generator asked for a normal sample that drew a skewed one must be labelled for the sample it drew.

Table 3. Deviations from the plan, with reasons.

8. A failed measurement, reported as failed

The plan requires a 0–2 interpretation rubric applied blind to system identity, with at least a fifth of reports double-scored and agreement reported. A first round was run with two independent raters. **It failed its reliability check.** Inter-rater agreement reached a kappa of -0.044 — worse than chance.

Diagnosis attributes this to the instrument, not to rater carelessness: the rubric admitted two defensible readings of the middle grade, and raters applied different ones consistently. The rubric has been rewritten with disjoint anchors and adjudicated calibration examples. A second round requires two human raters and is listed under remaining work.

The interpretation metric is therefore withheld rather than reported. Publishing a score from an instrument that demonstrably does not measure anything would be worse than reporting none, and the failure itself is a finding worth recording.

9. Remaining work

- **Execute the held-out evaluation.** The runner is resumable, validates the live path before starting, and refuses to begin work it cannot finish. Execution is the only outstanding step.
- **Score, bootstrap and analyse.** Every table regenerates from the raw logs by a single command, so this follows immediately once the runs are recorded.
- **Second blinded scoring round** with the rebuilt instrument, which requires two human raters.
- **Final report and packaging.** The report already regenerates from the logs and will incorporate the held-out section without manual editing.

10. Assessment

The project is on schedule against its gates. The principal risk remaining is not technical but one of resources: the held-out evaluation requires inference capacity, and the evaluation cannot be reported without it. Every control around that step — the frozen split, the frozen prompt hash, the resumable runner,

the execution guard — is in place and tested, so the step is bounded and repeatable rather than exploratory.

The second risk is interpretive. The development evidence is strong but optimistic by construction, and the held-out result may be weaker. The reporting structure is already built to state that outcome plainly if it occurs: the report distinguishes offline calibration, live preliminary and live held-out evidence throughout, and never blurs them.